Earners AI and ML by Aditya Jain sir - Lecture 23

```
n = [1,2,3,4,5]
```

```
# create a list to store the squares of each of the numbers in the list

sq = []
for elem in n:
  sq.append(elem**2)

print(sq)
```

```
[1, 4, 9, 16, 25]
```

```
#mentos zindagi - using functional programming ->

sq = list(map(lambda x: x**2, n))
sq
```

```
[1, 4, 9, 16, 25]
```

Lamda functions

```
# lambda function --> anonymous function
```

```
def sq(x):
  return x*x

print(sq(5))
```

```
25
```

```
sq = lambda x: x*x

# lambda parameters: expression
```

```
sq(6)
```

```
36
```

```
#write a lambda function to return the twice of the given number

twice = lambda x: 2*x

twice(4)
```

```
8
```

```
twice(6)
```

```
12
```

```
#Write a code to retur the sum of two given numbers


def add(a,b):
  return a+b
```

```
add(4,5)
```

```
9
```

```
add1 = lambda x,y: x+y
```

```
add1(4,6)
```

```
10
```

```
#Write a code to return the max of the two numbers

def mx(a,b):
  if a>b:
    return a
  else:
    return b
```

```
mx(4,5)
```

```
5
```

```
mx = lambda x,y: x if x>y else y

mx(4,7)
```

```
7
```

```
#Write a function to return if the given number is even or odd
def check(n):
  if n%2 == 0:
    return "Even"
  else:
    return "Odd"
```

```
check(5)
```

```
'Odd'
```

```
check(8)
```

```
'Even'
```

```
check1= lambda n: "Even" if n%2 == 0 else "Odd"
```

```
check1(4)
```

```
'Even'
```

```
check1(5)
```

```
'Odd'
```

```
#Write a lambda function to return the last character of the given string

last = lambda s: s[-1]
```

```
last("Aditya")
```

```
'a'
```

```
last("AJ")
```

```
'J'
```

```
#Write a code to return the product of the 3 numbers.
```

```
prod = lambda x,y,z : x*y*z
```

```
prod(4,5,6)
```

```
120
```

Map

```
# map(function, iterable) --> it returns a map object --> convert it into list --> list(map(..))
```

```
#Write a function to return a list of the cube of every number in the given list.

def cube(lst):
  cb = []
  for n in lst:
    cb.append(n**3)

  return cb
```

```
cube([5,7,3])
```

```
[125, 343, 27]
```

```
def cb(n):
  return n**3

cube = []
lst = [4,6,2]
for elem in lst:
  cube.append(cb(elem))

print(cube)
```

```
[64, 216, 8]
```

```
lst = [4,6,2]
cube = list(map(lambda x: x**3, lst))
print(cube)
```

```
[64, 216, 8]
```

```
cube = list(map(lambda x: x**3, [4,6,2]))
print(cube)
```

```
[64, 216, 8]
```

```
print(list(map(lambda x: x**3, [4,6,2])))
```

```
[64, 216, 8]
```

```
# write a code to return a list with the plus 5 of every number in the list.
```

```
def plus5(n):
  return n+5
```

```
lst = [5,7,2]
ans = []
for n in lst:
  ans.append(plus5(n))

print(ans)
```

```
[10, 12, 7]
```

```
lst = [5,7,2]

plus5 = list(map(lambda n: n+5, lst))

print(plus5)
```

```
[10, 12, 7]
```

```
#Given a list of strings, convert all the strings to all uppercase and display the new list
```

```
s = ["Aditya" ,  "Anshika", "Rahul" , "Raj"]

ups = []

for elem in s:
  ups.append(elem.upper())
```

```
print(ups)
```

```
['ADITYA', 'ANSHIKA', 'RAHUL', 'RAJ']
```

```
ss = ["Aditya" ,  "Anshika", "Rahul" , "Raj"]

ups = list(map(lambda s: s.upper(), ss))
print(ups)
```

```
['ADITYA', 'ANSHIKA', 'RAHUL', 'RAJ']
```

```
#Given a list of prices, return a new list with 18% gst included prices.

prices = [100,200,250, 400]
# expected: [118, 236, 295, 472]

gst_p = list(map(lambda x: 1.18*x, prices))
print(gst_p)
```

```
[118.0, 236.0, 295.0, 472.0]
```

Filter

```
num = [1,2,3,4,5,6,7]

#Write a code to create a list of all the even numbers of the given list

even = []

for n in num:
  if n%2 == 0:
    even.append(n)

print(even)
```

```
[2, 4, 6]
```

```
num = [1,2,3,4,5,6,7]

even = list(filter(lambda x: x%2 == 0, num))
print(even)
```

```
[2, 4, 6]
```

```
even = list(filter(lambda x: x%2 == 0, [1,2,3,4,5,6,7]))
print(even)
```

```
[2, 4, 6]
```

```
even = list(filter(lambda x: x%2 != 0, [1,2,3,4,5,6,7]))
print(even)
```

```
[1, 3, 5, 7]
```

Quiz

```
square = lambda x: x ** 2
```

```
print(square(6))
```

```
36
```

```
add = lambda a, b: a + b

numbers = [1, 2, 3]

result = list(map(lambda x: add(x, 5), numbers))

print(result)
```

```
[6, 7, 8]
```

```
numbers = [5, 10, 15, 20]

result = list(filter(lambda x: x > 10, numbers))

print(result)
```

```
[15, 20]
```

```
names = ["rahul", "amit", "neha"]

result = list(map(lambda x: x.capitalize(), names))

print(result)
```

```
['Rahul', 'Amit', 'Neha']
```

```
numbers = [12, 15, 18, 21, 24]

result = list(filter(lambda x: x % 6 == 0, numbers))

print(result)
```

```
[12, 18, 24]
```

```
numbers = [1, 2, 3]

result = list(map(lambda x: x + 10, numbers))

print(result)
```

```
[11, 12, 13]
```

```
numbers = [5, 8, 11, 14]

result = list(filter(lambda x: x % 2 != 0, numbers))

print(result)
```

```
[5, 11]
```

```
check = lambda x: "Adult" if x >= 18 else "Minor"

print(check(16))
```

```
Minor
```

```
words = ["AI", "Machine", "Python"]

result = list(map(len, words))

print(result)
```

```
[2, 7, 6]
```

DPP

```
numbers = [10, 15, 20]

result = list(map(lambda x: x // 5, numbers))

print(result)
```

```
numbers = [3, 6, 9, 12]

result = list(map(lambda x: x + 1, filter(lambda x: x % 6 == 0, numbers)))

print(result)
```

```
marks = [35, 45, 80, 25, 90]

passed = filter(lambda x: x >= 40, marks)

bonus = list(map(lambda x: x + 5, passed))

print(bonus)
```

```
numbers = [2, 3, 4]

result = list(
    map(
        lambda x: x ** 2,
        filter(lambda x: x > 2, numbers)
    )
)

print(result)
```

```
from functools import reduce

numbers = [1, 2, 3, 4, 5]

result= list(map(
        lambda x: x * 10,
        filter(lambda x: x % 2 != 0, numbers)
    )
)

print(result)
```